

1.6. Conceptual Schema And Logical Schema Design Phase

Under conceptual phase there is some terminology

1.6.1.Entity :An entity in a database is a thing, place, person or object that is independent of another. In databases, you store information about things in the real

world, and these things are database entities. Types:

A. Weak Entity: represented by double rectangle.

B. Strong Entity: represented by single rectangle.

1.6.2. Attribute: Attribute is a piece of data that describes an entity. Types:

A. simple (atomic) vs. composite entity: Simple contains single value while Composite divide into sub parts. B. Single valued vs. multi valued attribute Single value: have single value and represented by oval shape. Multi value: have more than one shape and represented by double oval.

C. Stored vs. Derived attribute: Stored: not possible to divide. Derived: The value may be derived, represented by dot line oval. D. Null values: The attribute which are not applicable but the value and key attribute cannot be null. E. Default value: Assumed value if not exist value or unknown.

1.6.3. Relationship : a situation that exist between two relational database tables when one table has a foreign key that reference the primary key of the other table.

1.6.4. Identify Entity

- | | |
|--------------|------------|
| 1. Passenger | 5.seat |
| 2. Airline | 6.flight |
| 3. Booking | 7. user |
| 4. Payment | 8.auditlog |

Identify attribute

➤ Passenger

-passid
-Country
-Passport_number
-Name
-Phone_number
-Created_at

➤ Airline

-airid
-Country

flight

-flightid-
-arrival_time
-departure_time
-flightcode
- origin
- destination

➤ **Booking**

-bookid
-Status
-Booking_date

➤ **Payment**

-Paymentid
-Status
-Payment_date
-Method

➤ **Seat**

-Seatid
-Status
-Seat_number

➤ **User**

-Userid
-Username
-Password_hash

audit_log

-auditid
-action
-timestamp

